Homework 1

Abbie Brunner

February 20, 2026

Contents

Problem 1

Greens Function Validation

We were asked to prove that $G = \frac{\ln(r)}{2\pi}$ is a valid Greens function. There are two properties that need to be satisfied being

$$\nabla^2 G = 0, \qquad \text{and} \qquad \lim_{r\to 0} \iint \nabla^2 G \, \mathrm{d}\mathbf{x} = \lim_{r\to 0} \iint \delta(\mathbf{y} - \mathbf{x}) = 1$$

Proving the first condition is fairly straightforward:

$$\begin{aligned}
\nabla^2 G &= \frac{\partial^2 G}{\partial r^2} + \frac{1}{r}\frac{\partial G}{\partial r} + \frac{1}{r^2}\frac{\partial^2 G}{\partial \theta^2}; \quad r = (y_i - x_i)(y_i - x_i) \\
&= \frac{\partial^2}{\partial r^2}\left(\frac{\ln(r)}{2\pi}\right) + \frac{1}{r}\frac{\partial}{\partial r}\left(\frac{\ln(r)}{2\pi}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}\left(\frac{\ln(r)}{2\pi}\right) \\
&= \frac{\partial}{\partial r}\left(\frac{1}{2\pi r}\right) + \frac{1}{r}\left(\frac{1}{2\pi r}\right) + 0 \\
&= -\frac{1}{2\pi r^2} + \frac{1}{2\pi r^2} \\
&= 0
\end{aligned}$$

The second condition requires the use of Stokes' theorem. Transforming this into a path integral allows us to simplify the integral and solve proving that the second condition is satisfied.

$$
\begin{aligned}
\iint \nabla^2 G \,\mathrm{d}\mathbf{x} &= \oint \nabla G \cdot \hat{\mathbf{n}} \,\mathrm{d}S \\
&= \oint \frac{\partial G}{\partial r} r \,\mathrm{d}\theta \\
&= \int_0^{2\pi} \frac{r}{2\pi r} \\
&= \frac{1}{2\pi} \int_0^{2\pi} \mathrm{d}\theta \\
&= 1
\end{aligned}
$$

Problem 2

Boundary Element Implementation

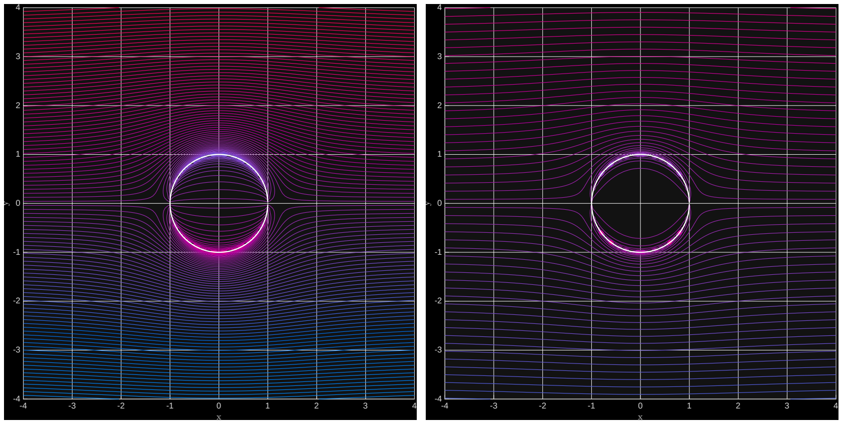
The condition presented here is uniform flow around an object. Using the Green's function from the last portion to solve the 2D Laplace equation we solve for the Streamfunction that makes up the ψ_1 portion of the combined ψ. We can use the boundary condition at the surface of $\psi = 0$ and thus $\psi_1 = -\psi_0$ to evaluate Ψ at each panel from the matrix approximation of

$$\frac{1}{2}\psi_1(\mathbf{y}) = \oint \psi \frac{\partial G}{\partial n} - G\frac{\partial \psi_1}{\partial n}\,\mathrm{d}s$$

for ψ at the surface. Solving for $\frac{\partial \Psi}{\partial n}$ is then done by inverting the matrix B. From there ψ_1 can be added in superposition with ψ_0 to evaluate for the flow away from the cylinder surface

$$\underbrace{\begin{bmatrix} -0.5 & \frac{\partial G_{12}}{\partial n_2}\partial s_2 & \frac{\partial G_{13}}{\partial n_3}\partial s_3 \dots & \\ \frac{\partial G_{21}}{\partial n_1}\partial s_1 & -0.5 & & \vdots \\ \vdots & & \ddots & \vdots \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} \psi_1 \\ \psi_2 \\ \vdots \end{bmatrix}}_{\Psi} = \underbrace{\begin{bmatrix} 0 & G_{12} & G_{13} & \dots \\ G_{21} & 0 & G_{23} & \vdots \\ \vdots & \dots & \dots & \ddots \end{bmatrix}}_{B} \underbrace{\begin{bmatrix} \left.\frac{\partial \psi_1}{\partial n}\right|_1 \\ \left.\frac{\partial \psi_1}{\partial n}\right|_2 \\ \vdots \end{bmatrix}}_{\frac{\partial \Psi}{\partial n}}$$

Using the streamfunction that we computed we can compute a contour as shown in fig. 2.1. Between the two cylinders we see that with a higher number of panels there is a significantly higher resolution at the boundaries which is



(a) Cylinder with 100 panels (b) Cylinder with 200 panels

Figure 2.1: BEM Cylinders

expected. On both cylinders there are some interesting artifacts at the surface, but I am unsure what those are. I suspect a tolerance error somewhere is to blame, but I need to do more exploring of the Boundary Element Method to track down what that is.

A comparison of this against the exact solution

$$\psi(r,\theta) = \left(r - \frac{1}{r}\right)\sin(\theta)$$

shows fairly good.

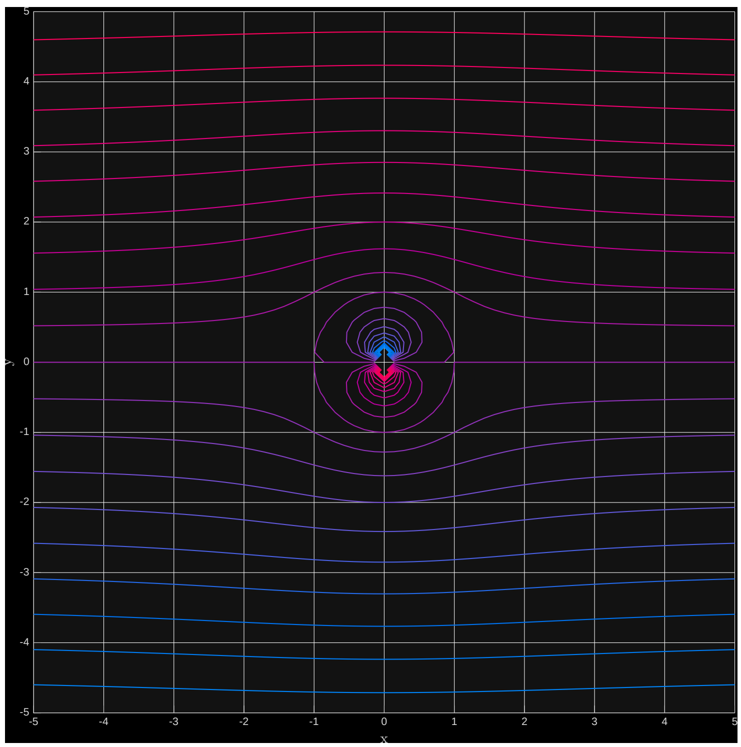


Figure 2.2: Exact Solution

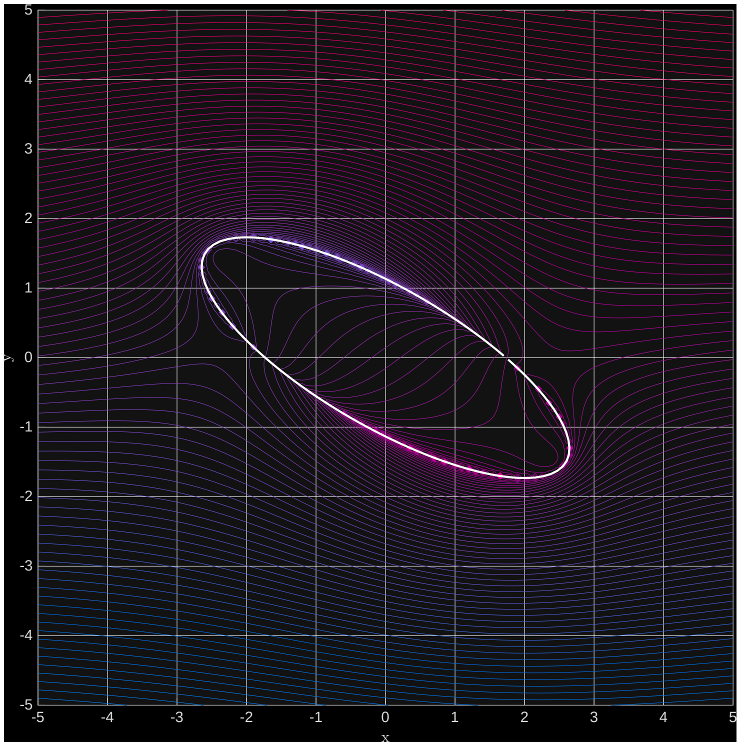


Figure 2.3: 3:1 Ellipse, α

The final part of this problem asked for flow around an ellipse by leveraging BEMs ability to work around arbitrary geometries. To do this the polar description of an ellipse was used as follows:

$$r(\theta) = \frac{1}{1 - \sqrt{(e\cos(\theta))^2}}, \qquad \text{with} \qquad e = \sqrt{1 - \frac{b^2}{a^2}}$$

The boundary can the be set using the relations $x = r\cos(\theta)$ and $y = r\sin(\theta)$ and then most of the method remains the same. Care needs to be taken with the normal vectors for each panel. This was done with a simple approximation taking the vector between the two points defining each panel and finding the orthogonal normal vector. The magnitude of that vector was used for the panel length.

Appendix A

Analytic Flow

```
close all;
clear variables

Nr = 100;
r_max = 10;

N = 100;
dth = 2*pi/N;
th = [dth/2:dth:2*pi-dth/2];

dr = (r_max-1)/(Nr-1);
[rTest thetaTest] = ndgrid(1:dr:r_max, 0:dth:2*pi-dth);
psi_polar = (rTest-1./rTest).*sin(thetaTest);
x = rTest.*cos(thetaTest);
y = rTest.*sin(thetaTest);
psiFun = @(x,y) y-(y./(x.^2+y.^2));
psiexact = psiFun(x,y);

part4 = figure;
hold on;
psiHandle = fcontour(psiFun,[-5 5 -5 5],LineWidth=1);
psiHandle.LevelList = linspace(-5,5,21);
colormap(nebula)
grid on
pbaspect([1 1 1]);
xlabel('x',Interpreter='latex',Fontsize=14)
ylabel('y',Interpreter='latex',Fontsize=14)
saveas(part4,'part4el','epsc')
```

Appendix B

Cylinder Flow

```
function [] = laplacian2DCyl(N)
% Boundary Element Method (BEM) for plane-wave scattering from cylinder
%
% Joseph W. Nichols, 2024

%k = 2*pi/lambda;  % wavenumber

% circle
dth = 2*pi/N;
th = [dth/2:dth:2*pi-dth/2];
x = [cos(th); sin(th)];   % collocation points

n = -x;                   % normal vectors (pointing into the cylinder)
ds = dth*ones(N);         % panel sizes


% Boundary element matrices
A = zeros(N,N);
B = zeros(N,N);
for i2 = 1:N
    for i1 = 1:N
        if (i1 == i2)
            A(i1,i2) = -0.5;
            B(i1,i2) = 0.0;
        else
            r = x(:,i2) - x(:,i1);  % source - observer
            rmag = norm(r);
            r = r/rmag;
            %G    = -1i/4*besselh(0,k*rmag);
            %dGdr =  1i*k/4*besselh(1,k*rmag);
```

```
            G = log(rmag)/2*pi;
            dGdr = 1/2/rmag/pi;
            A(i1,i2) = ds(i2)*dot(r,n(:,i2))*dGdr;
            B(i1,i2) = ds(i2)*G;
        end
    end
end

%drhos0 = 1i*k*x(1,:)'.*exp(1i*k*x(1,:)');  % drho/dn of scattered wave
     on cyl surf
%drhos0 = 1./2./pi./x(1,:)'+x(1,:)'.*cos(th)';
%rhos0  = A\(B*drhos0);                     % where the magic happens
psi1 = -sin(th)';
dpsi = (A*psi1)\B;

part2 = figure;
subplot(211);
%hold on;
plot(th, psi1);
grid on;
xlim([0 6])
ylim([-1.5 1.5])
xlabel('$\theta$',Interpreter='latex',Fontsize=14)
ylabel('$\psi_1$',Interpreter='latex',Fontsize=14)
subplot(212);
%hold on;
plot(th,dpsi);
grid on;
xlabel('$\theta$',Interpreter='latex',Fontsize=14)
ylabel('$\frac{\delta\psi_1}{\delta n}$',Interpreter='latex',Fontsize
    =14)
saveas(part2,'part2cyl','epsc')


% Visualize solution
[XX,YY] = meshgrid(-4:.05:4, -4:.05:4);
RR = sqrt(XX.^2+YY.^2);
TT = atan(YY./XX);
PsiAnalytic = (RR - 1./RR)*sin(TT);

%Nr = 100;
%
%r_max = 10;
%
%dr = (r_max-1)/(Nr-1);
%[rTest thetaTest] = ndgrid(1:dr:r_max, 0:dth:2*pi-dth);
```

```
%psi_polar = (rTest-1./rTest).*sin(thetaTest);
%x = rTest.*cos(thetaTest);
%y = rTest.*sin(thetaTest);
%psiFun = @(x,y) y-(y./(x.^2+y.^2));
%psiexact = psiFun(x,y);
%psiHandle = fcontour(psiFun,[-5 5 -5 5],LineWidth=1);
%
%part4 = figure;
%hold on;
%psiHandle.LevelList = linspace(-5,5,21);
%grid on
%pbaspect([1 1 1]);
%xlabel('x',Interpreter='latex',Fontsize=14)
%ylabel('y',Interpreter='latex',Fontsize=14)
%saveas(part4,'part4el','epsc')

%rhos = 0*XX;

psi = 0*XX;

for i1 = 1:N  % loop over source panels

    Rx = x(1,i1) - XX;    % source - observer(s)
    Ry = x(2,i1) - YY;
    rmag = sqrt(Rx.^2 + Ry.^2);
    Rx = Rx./rmag;
    Ry = Ry./rmag;
    %G    = -1i/4*besselh(0,k*rmag);
    %dGdr =  1i*k/4*besselh(1,k*rmag);
    G = log(rmag)./2.*pi;
    dGdr = 1./2./rmag./pi;
    psi = psi + ds(i1)*(psi1(i1).*dGdr-G.*dpsi(i1));
    %rhos = rhos + ds(i1)*(rhos0(i1)*(Rx*n(1,i1) + Ry*n(2,i1)).*dGdr -
        drhos0(i1)*G);
end

%psi = psi + RR.*sin(TT);
psi = psi + YY;


part3 = figure;
hold on;
contour(XX, YY, psi, 100, Linewidth=1);
colormap(nebula)
grid on;
plot(x(1,:), x(2,:),color='w',LineWidth=2);
```

```
pbaspect([1 1 1]);
xlabel('x',Interpreter='latex',Fontsize=14)
ylabel('y',Interpreter='latex',Fontsize=14)
set(gca,'fontsize', 14);
saveas(part3,'part3cyl' + string(N),'epsc')

%
%% incident wave
%rhoi = exp(1i*k*XX);
%
%% Animate solution in time
%for i1 = 1:100
%    pcolor(XX,YY,real((rhos)*exp(-1i*2*pi*i1/20)));
%%    pcolor(XX,YY,real((rhoi + rhos)*exp(-1i*2*pi*i1/20)));
%    shading interp;
%    colorbar;
%    caxis([-1 1]);
%    axis equal;
%    drawnow;
%end

end
```

Appendix C

Ellipse Flow

```
function [] = laplacian2DOval(N)
% Boundary Element Method (BEM) for plane-wave scattering from cylinder
%
% Joseph W. Nichols, 2024

%k = 2*pi/lambda;  % wavenumber

% circle
dth = 2*pi/N;
th = [dth/2:dth:2*pi-dth/2];
%x = [cos(th); sin(th)];   % collocation points

%n = -x;                   % normal vectors (pointing into the cylinder
    )
%ds = dth*ones(N);         % panel sizes

e = sqrt(1-(1/3)^2);

%r = 1./sqrt(1-(e.*cos(th)).^2);
r = 1./sqrt(1-(e.*cos(th+pi/6.0)).^2);

x = [r.*cos(th); r.*sin(th)];

%size(x)

n = 0*x;
ds = 0*dth;

for lv1 = 2:N
   p = x(:, lv1) - x(:,lv1-1);
```

```
   n(:,lv1-1) = -p;
   ds(lv1-1) = norm(p);
end

n(:,size(x)) = (x(:, size(x)) - x(:,1));
ds(size(x)) = norm(x(:, size(x)) - x(:,1));

% Boundary element matrices
A = zeros(N,N);
B = zeros(N,N);
for i2 = 1:N
    for i1 = 1:N
        if (i1 == i2)
            A(i1,i2) = -0.5;
            B(i1,i2) = 0.0;
        else
            r = x(:,i2) - x(:,i1);  % source - observer
            rmag = norm(r);
            r = r/rmag;
            %G    = -1i/4*besselh(0,k*rmag);
            %dGdr =  1i*k/4*besselh(1,k*rmag);
            G = log(rmag)/2*pi;
            dGdr = 1/2/rmag/pi;
            A(i1,i2) = ds(i2)*dot(r,n(:,i2))*dGdr;
            B(i1,i2) = ds(i2)*G;
        end
    end
end

%drhos0 = 1i*k*x(1,:)'.*exp(1i*k*x(1,:)');  % drho/dn of scattered wave
     on cyl surf
%drhos0 = 1./2./pi./x(1,:)'+x(1,:)'.*cos(th)';
%rhos0  = A\(B*drhos0);                      % where the magic happens
psi1 = -sin(th)';
dpsi = (A*psi1)\B;

part2 = figure;
subplot(211);
%hold on;
plot(th, psi1);
grid on;
xlim([0 6])
ylim([-1.5 1.5])
xlabel('$\theta$',Interpreter='latex',Fontsize=14)
ylabel('$\psi_1$',Interpreter='latex',Fontsize=14)
subplot(212);
```

```
%hold on;
plot(th,dpsi);
grid on;
xlabel('$\theta$',Interpreter='latex',Fontsize=14)
ylabel('$\frac{\delta\psi_1}{\delta n}$',Interpreter='latex',Fontsize
    =14)
saveas(part2,'part2el','epsc')


% Visualize solution
[XX,YY] = meshgrid(-5:.05:5, -5:.05:5);
RR = sqrt(XX.^2+YY.^2);
TT = atan(YY./XX);
%rhos = 0*XX;

psi = 0*XX;

for i1 = 1:N  % loop over source panels

    Rx = x(1,i1) - XX;    % source - observer(s)
    Ry = x(2,i1) - YY;
    rmag = sqrt(Rx.^2 + Ry.^2);
    Rx = Rx./rmag;
    Ry = Ry./rmag;
    %G    = -1i/4*besselh(0,k*rmag);
    %dGdr =  1i*k/4*besselh(1,k*rmag);
    G = log(rmag)./2.*pi;
    dGdr = 1./2./rmag./pi;
    psi = psi + ds(i1)*(psi1(i1).*dGdr-G.*dpsi(i1));
    %rhos = rhos + ds(i1)*(rhos0(i1)*(Rx*n(1,i1) + Ry*n(2,i1)).*dGdr -
        drhos0(i1)*G);
end

%psi = psi + RR.*sin(TT);
psi = psi + YY;
%psi = psi + exp(-i*(pi/6));

part3 = figure;
hold on;
%contour(XX, YY, psi, 100);
contour(XX, YY, real(psi), 100);
colormap(nebula)
grid on;
plot(x(1,:), x(2,:),color='w',LineWidth=2);
pbaspect([1 1 1]);
xlabel('x',Interpreter='latex',Fontsize=14)
```

```
ylabel('y',Interpreter='latex',Fontsize=14)
set(gca,'fontsize', 14);
saveas(part3,'part3el','epsc')
%
%% incident wave
%rhoi = exp(1i*k*XX);
%
%% Animate solution in time
%for i1 = 1:100
%    pcolor(XX,YY,real((rhos)*exp(-1i*2*pi*i1/20)));
%%     pcolor(XX,YY,real((rhoi + rhos)*exp(-1i*2*pi*i1/20)));
%    shading interp;
%    colorbar;
%    caxis([-1 1]);
%    axis equal;
%    drawnow;
%end

end
```